

Learning Resource On Speech and Video Processing

Module-1: Speech Processing Concepts Chapter-3: Time Frequency Analysis of Speech Signal

**Prepared By:
Dr. Kunal Anand, Asst. Professor
SCE, KIIT, DU, Bhubaneswar-24**

Chapter Outcomes

- After completing this module, students will be able to:
 - Explain the concept of time-frequency representation and spectrograms for speech analysis.
 - Interpret speech characteristics using spectrograms.
 - Apply convolution to analyze simple speech processing systems.
 - Explain pole-zero models and their significance in speech system representation.
 - Relate spectrograms, convolution, and pole-zero models to real-world speech processing applications.

Time-domain representaion of Speech Signal

- A **time-domain** representation of a speech signal is the simplest way to represent speech.
- It shows how the **amplitude** (air pressure variation) of the speech signal **changes with time**.
- A **time-domain** representation of a speech signal is a graph that consists of following two axes:
 - **X-axis: Time (seconds or milliseconds)**
 - Represents the time over which the speech signal is observed.
 - The **duration** of the signal can be determined directly from this axis.
 - Indicates when different speech events occur, such as the beginning of a word, pauses, and the end of speech.
 - Measured in **seconds (s) or milliseconds (ms)**.

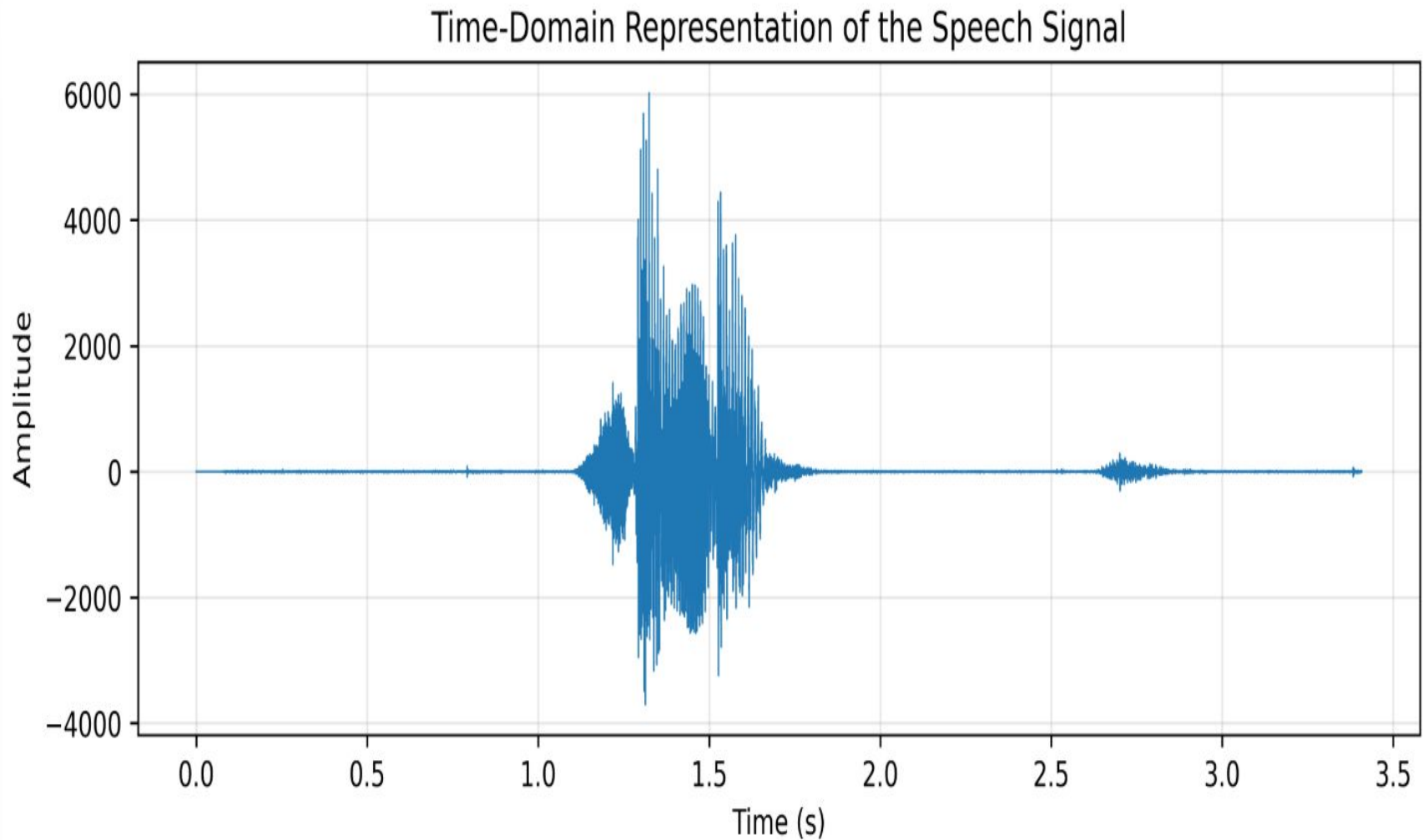
–Y-axis: Amplitude (sound pressure or voltage)

- Represents the instantaneous **amplitude** of the speech signal.
- Indicates the **strength or intensity** of the sound wave at each instant.
- Measured as:
 - Voltage (V) in analog systems,
 - Normalized amplitude (-1 to $+1$) in digital systems, or
 - Sound pressure (Pa) in acoustic measurements.
- Positive amplitude represents positive air pressure variation whereas, negative amplitude represents negative air pressure variation.
- Larger amplitude indicates louder speech whereas, smaller amplitude represents softer speech or silence.

Sample Example “Sunday”



Time-domain representation



- **Interpretation**

- **X-axis (Time duration in seconds):** The word “**sunday**” lasts for approximately 3.5 seconds.
- **Y-axis (Amplitude):** The amplitude ranges approximately from -4000 to 6000.
- **0 to 1.1 s:** Initial Silence (Nearly flat showing no speech activity)
- **1.1 to 1.7:** Production of speech sound
- **1.7 – 2.6 s:** Silence; The waveform returns close to zero, representing a pause or the end of the first spoken segment.
- **2.6 – 2.9 s:** Low-Amplitude Speech; A smaller waveform appears, indicating a short, softer speech segment
- **2.9–3.5 seconds:** Ending Silence; the waveform becomes nearly flat again, indicating the end of the recording.

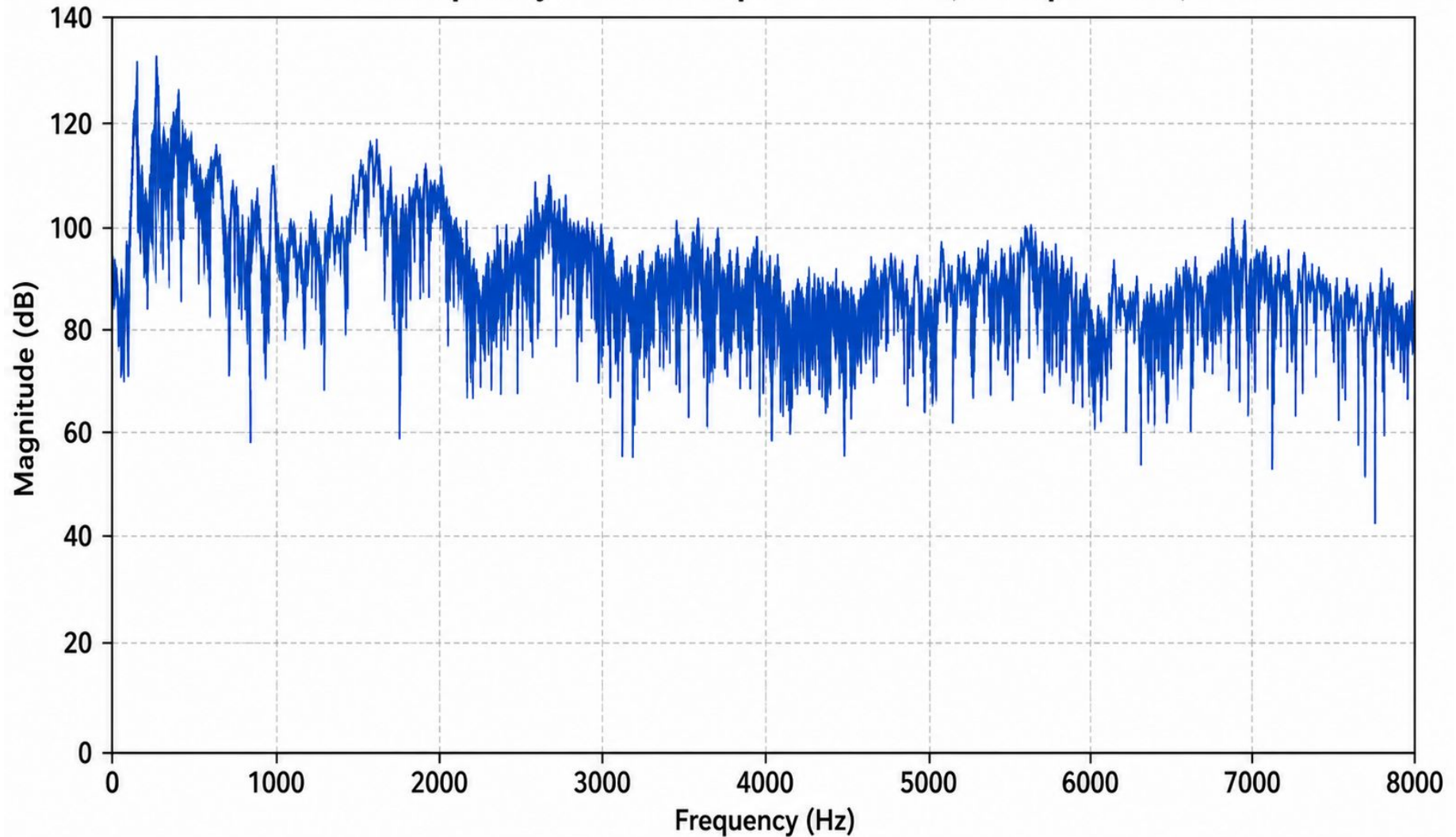
Frequency-domain representation of Speech Signal

- The **frequency-domain** representation of a speech signal describes how the **energy** of the speech signal is distributed among **different frequencies** rather than showing how the signal changes with time.
- A speech signal is composed of many sinusoidal components with different **frequencies, amplitudes, and phases**.
- By applying the **Fourier Transform (FT)** or **Fast Fourier Transform (FFT)** to a time-domain speech signal, these frequency components can be identified and analyzed.
- The frequency-domain representation answers "**Which frequencies are present in the speech signal and how strong are they?**"

- A **frequency-domain** graph consists of two axes:
 - **X-axis: Frequency (Hz)**
 - Represents the **frequency** components of speech.
 - Measured in **Hertz (Hz)**.
 - Typically ranges from **0 Hz to the Nyquist frequency**.
 - **Y-axis: Magnitude (Amplitude or dB)**
 - Represents the **strength (energy)** of each frequency component.
 - Can be displayed as: Linear amplitude, Power, Magnitude in decibels (dB)
- **Higher peaks indicate stronger frequency components.**

contd...

Frequency-Domain Representation (FFT Spectrum)



- **Interpretation**

- **X-axis** (Frequency in Hz):

- Represents the **frequency components** present in the speech signal ranging from **0 to 8000Hz**.
 - This means that the speech was sampled at **16 kHz**, giving a **Nyquist frequency of 8 kHz**.

- **Y-axis** (Magnitude in dB):

- Represents the **strength or energy** of each frequency component. Here, it lies between **40dB to 140dB**.
 - Higher magnitude values indicate stronger frequency components.

- **Interpretation**

- **0–1000 Hz:** Most of the signal **energy** is concentrated here, representing the fundamental frequency (pitch) and lower harmonics.
 - **1000–4000 Hz:** This region contains **formants**, which are important for identifying **vowels** and understanding speech.
 - **4000–8000 Hz:** The magnitude gradually decreases, but this range still carries **consonant** and **fricative** information, improving speech clarity.
-
- **Relationship with the given Time-Domain Waveform:**
 - The frequency spectrum is derived from the time-domain speech signal, showing how the signal's energy is distributed across different frequencies.

Time-domain VS Frequency-domain

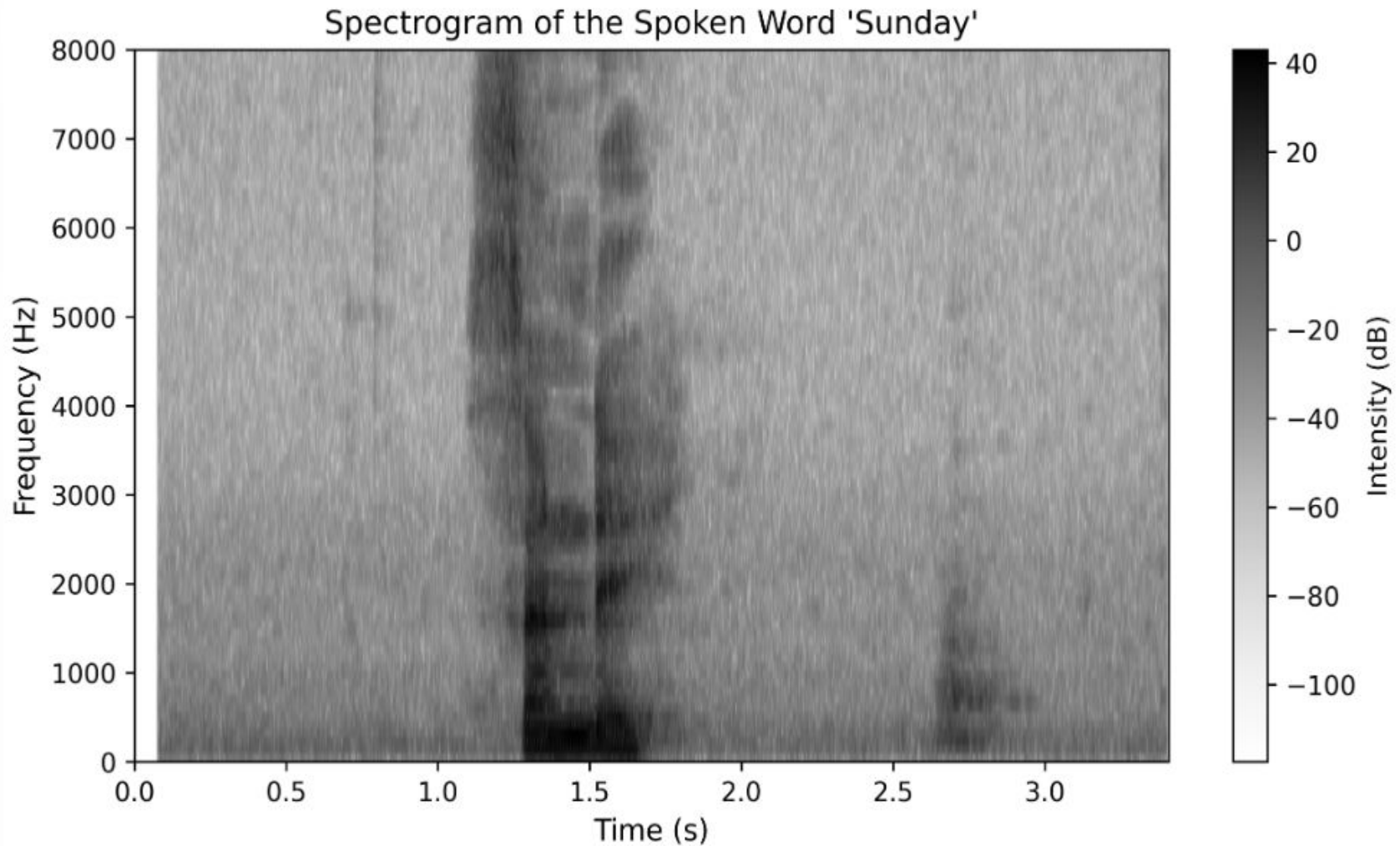
Time-domain	Frequency-domain
Shows how the amplitude of the signal varies with time	Shows how the speech's energy is distributed across different frequencies
X-axis represents time (s/ms) while Y-axis represents Amplitude	X-axis indicates frequency (Hz) while Y-axis represents magnitude or power (dB)
Provides information like loudness, pauses, and speech timing	Tells about pitch, harmonics, formants, and spectral energy distribution
Can be obtained directly from the recorded speech signal	Can be obtained by applying FT or FFT on the time-domain signal
Application areas: Speech segmentation, silence detection	Application areas: Speech recognition, speaker recognition, pitch estimation, ...

- A **spectrogram** is a **time-frequency** representation of a speech signal that shows how the **frequency** content of the signal **changes over time**.
- Unlike the **time-domain** representation, which shows **amplitude versus time**, and the **frequency-domain** representation, which shows **magnitude versus frequency**, a **spectrogram** combines **both types** of information in a single plot.
- It is generated by **dividing** the speech signal into **short overlapping frames** and applying the **Short-Time Fourier Transform (STFT)** to each frame.

Why Spectrogram???

- A **spectrogram** combines the advantages of both time- and frequency-domain representations:
 - Shows when a speech sound occurs (**time information**).
 - Shows which frequencies are present (**frequency information**).
 - Displays the **movement of formants over time**.
 - Helps distinguish **voiced** and **unvoiced sounds**.
 - Reveals transitions between phonemes and syllables.
- **Spectrogram** is highly useful in applications like speech recognition, speaker identification, speech synthesis, phoneme segmentation, formant analysis, emotion recognition, audio enhancement and noise reduction.

Spectrogram for the speech “Sunday”



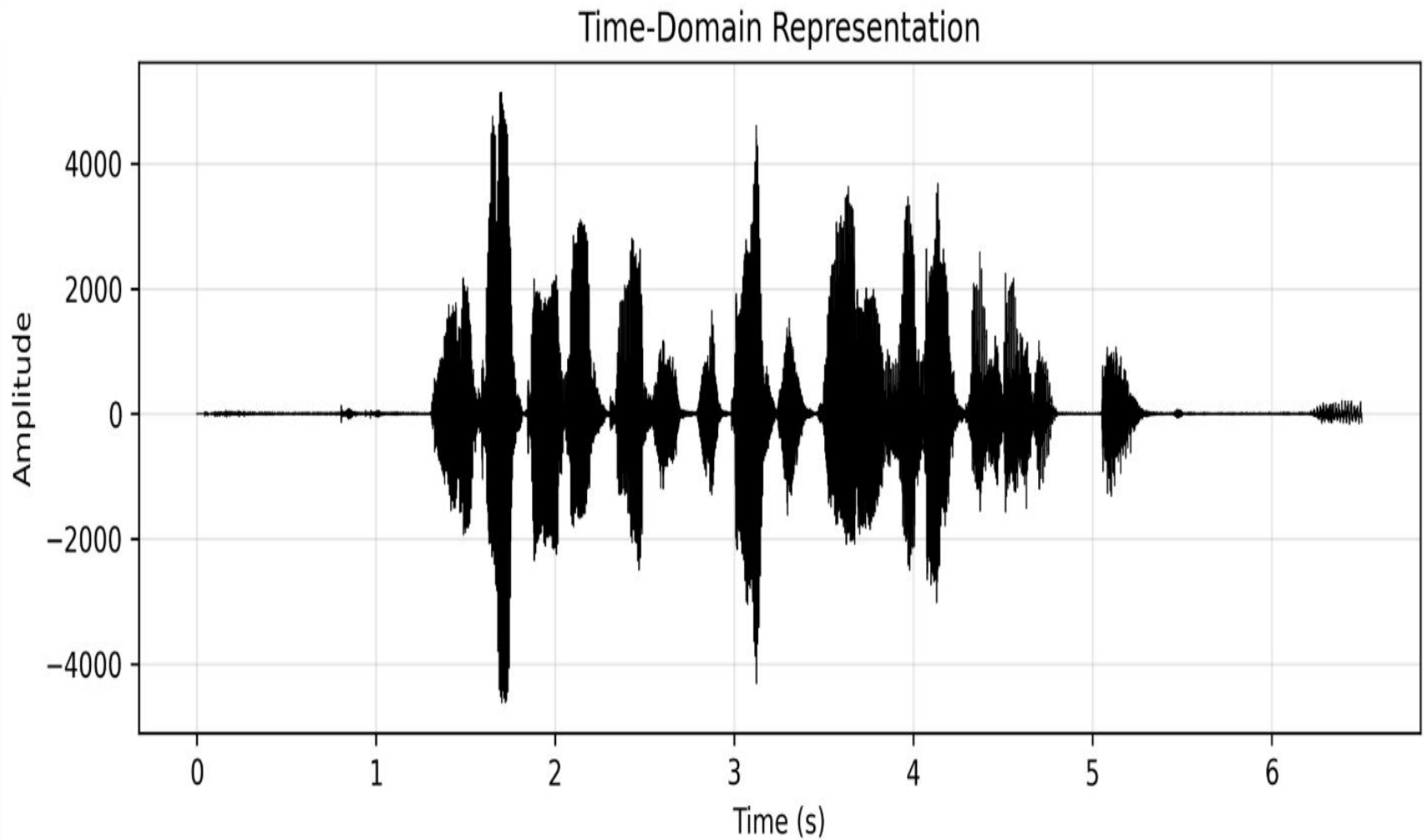
- Spectrogram for the speech “Sunday”
 - **X-axis:** duration of the signal (approx. 3.4 s)
 - **Y-axis:** frequency component (0 to 8000 Hz)
 - **Color Intensity (dB):** Indicates the energy of the speech signal at each time and frequency. **Darker** regions represent **higher energy**, while **lighter** regions represent **lower energy**.
- **Interpretation:**
 - **0–1.2 seconds:** Initial Silence indicating very little speech energy.
 - **1.2–1.8 seconds: Main Speech Segment**
 - Dark vertical and horizontal patterns appear
 - Most of the energy is concentrated below 3 kHz

- The horizontal dark bands represent **formants (F1, F2, and F3)**, which are the resonance frequencies of the vocal tract and are important for vowel identification.
- **2.7–2.9 seconds: Short Speech Segment**
 - A smaller dark region appears, indicating another brief speech sound or low-energy utterance.
 - **Remaining Duration**
 - The spectrogram returns to a lighter shade, indicating silence or background noise.

Sample Speech “Welcome to the course Speech and Video Processing”

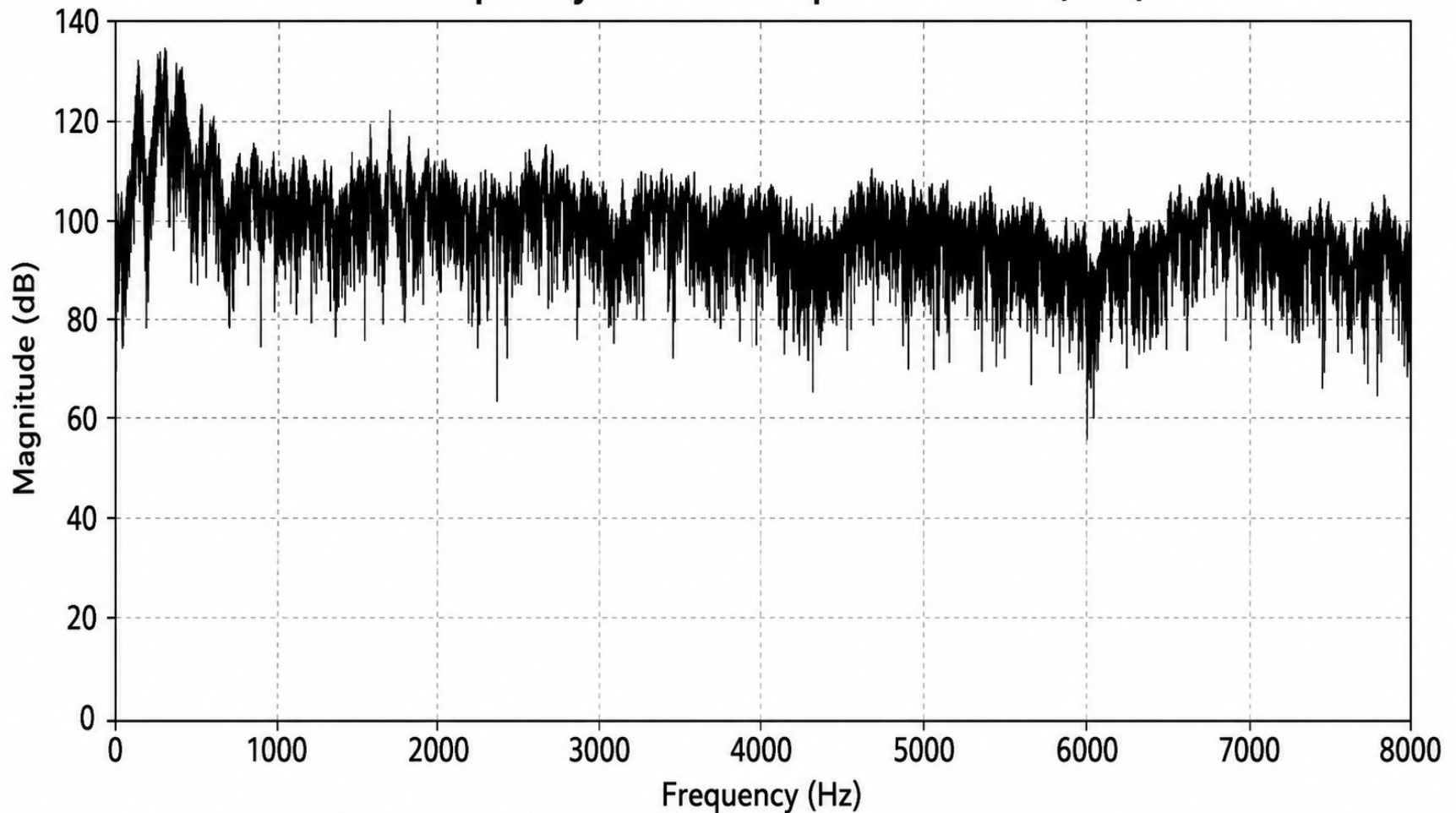


Time-domain representation

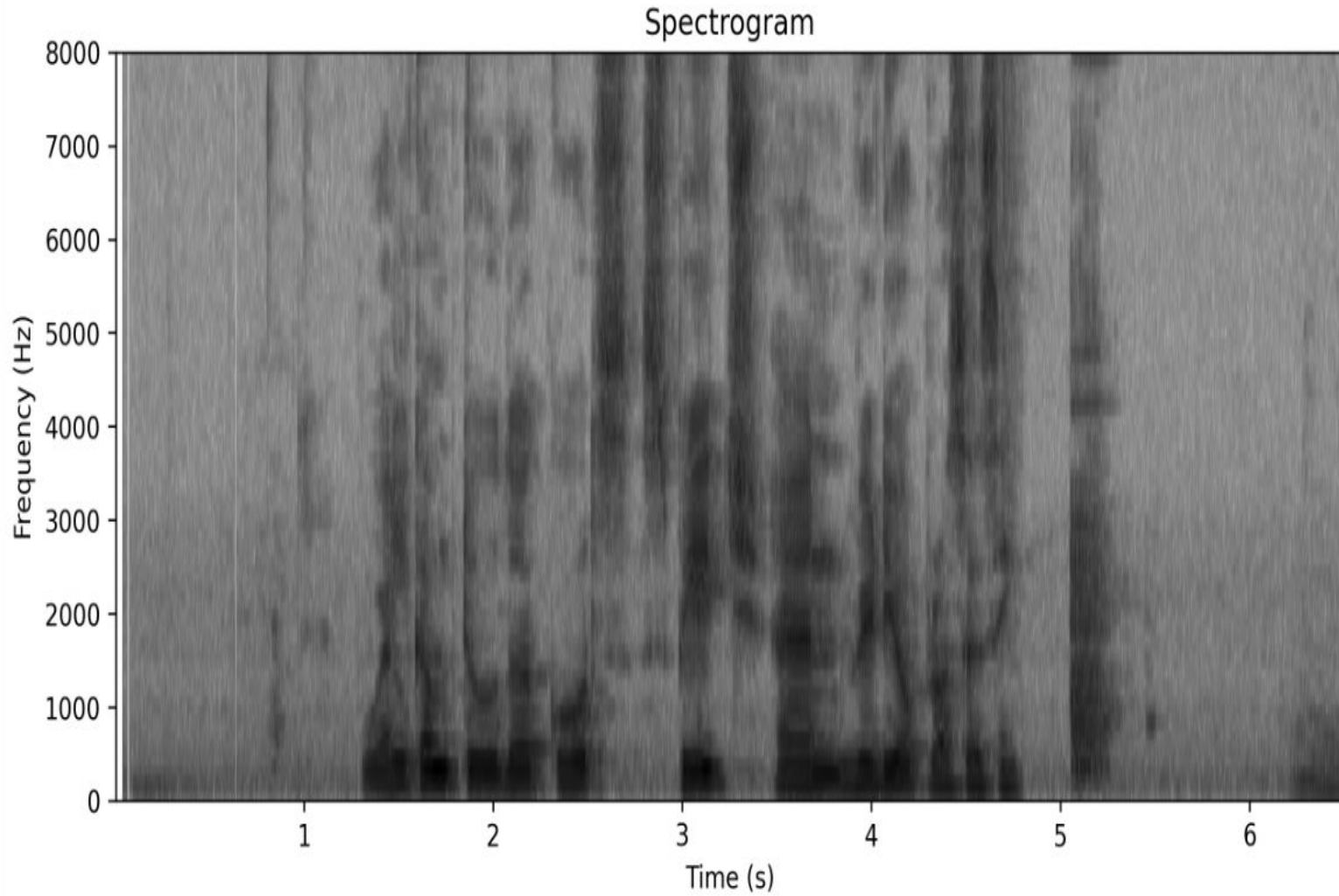


Frequency-domain representation

Frequency-Domain Representation (FFT)



Spectrogram



Discrete Time Systems

- A *discrete-time system* is a system that transforms an input sequence ($x[n]$) to an output sequence $y[n]$,

$$y[n] = T(x[n])$$

- **Example:** Microphone $\rightarrow$ converts acoustic speech into electrical signal.
- In **speech processing**, discrete-time systems model the vocal tract filter.

Linear Time Invariant (LTI) Systems

- A **Linear Time-Invariant (LTI)** system is a system that produces an **output signal** from an **input signal** while satisfying two important properties:
 - **Linearity**
 - **Time Invariance**
- **LTI systems** are very important in speech processing because the human vocal tract, microphones, and communication channels can often be approximated as LTI systems.
- An **LTI system** can be represented as below:

$$y[n] = T(x[n])$$

where,

T represents the system operation

Properties of LTI system

- An **LTI system** must have following two properties:
- **Linearity:** A system is **linear** if it follows the **scalability** and **additivity** property.
 - **Scalability:**
 - if $y(n) = x(n)$ then $ay(n) = ax(n)$
 - If the input is increased by a value then the output will also be increased by the same value.
 - **Additivity:**
 - if $y_1(n) = x_1(n)$ and $y_2(n) = x_2(n)$ then $y_1(n) + y_2(n) = x_1(n) + x_2(n)$
 - The response to multiple inputs is equal to the sum of individual responses.

- **Time Invariance:** A system is time invariant if its behavior does not change with time.
- if $\mathbf{y}(\mathbf{n}) = \mathbf{x}(\mathbf{n})$ then for a delayed input $\mathbf{x}(\mathbf{n} - \mathbf{n}_0)$ the output should be $\mathbf{y}(\mathbf{n} - \mathbf{n}_0)$; **where $\mathbf{n}_0 > 0$.**
- In other words, the system gives the same output whether the input is applied today or tomorrow.
- The signal's shape remains same, only the time location changes.

Sample Example

- Let's consider the following transformation

$$y[n] = x[n] + x[n - 1]$$

Check if the given system is time invariant or not.

- Input:**

$$x[n] = \{1, 2, 3\}$$

- System output:**

$$y[n] = \{1 + 0, 2 + 1, 3 + 2\} = \{1, 3, 5\}$$

- Shift input by $n_0 = 1$** ; the signal has been shifted one step to the right. So, the shifted input is now $\{0, 1, 2\}$

- R. H. S.: Output of shifted input:**

$$\begin{aligned} T(x[n - 1]) &= T(\{0, 1, 2\}) = \{0 + 0, 1 + 0, 2 + 1\} \\ &= \{0, 1, 3\} \end{aligned}$$

Linearity: Principle of Superposition

- The **principle of superposition** says that “The response to a **combination of inputs** is equal to the **combination of the individual responses**.”
- Let $x_1[n]$ and $x_2[n]$ be the **inputs** to a discrete-time system that produces $y_1[n]$ and $y_2[n]$ as **outputs** respectively.
- For arbitrary constants a and b , the system is **linear** if and only if

$$T(ax_1[n] + bx_2[n]) = aT(x_1[n]) + bT(x_2[n])$$

OR

$$T(ax_1[n] + bx_2[n]) = ay_1[n] + by_2[n]$$

- which is sometimes referred to as the “**principle of superposition**”.

Sample Example

Let's consider the following transformation

$$\mathbf{y}[n] = \mathbf{T}(\mathbf{x}[n]) = \mathbf{2} * \mathbf{x}[n]$$

- **Input 1:** $x_1[n] = \{1, 2\} \rightarrow$ **Output:** $y_1[n] = 2 * x_1[n] = \{2, 4\}$
- **Input 2:** $x_2[n] = \{3, 4\} \rightarrow$ **Output:** $y_2[n] = 2 * x_2[n] = \{6, 8\}$
- Let's take **a = 3, b = 2**
- **L. H. S.**

$$\begin{aligned} \mathbf{T}(\mathbf{3x}_1[\mathbf{n}] + \mathbf{2x}_2[\mathbf{n}]) &= \mathbf{T}\{(3 * 1 + 3 * 2), (2 * 3 + 2 * 4)\} \\ &= \mathbf{T}(\{9, 14\}) = \{\mathbf{18}, \mathbf{28}\} \end{aligned}$$

- **R. H. S.**

$$\begin{aligned} \mathbf{3} * \mathbf{y}_1[\mathbf{n}] + \mathbf{2} * \mathbf{y}_2[\mathbf{n}] &= 3 * \{2, 4\} + 2 * \{6, 8\} \\ &= \{6, 12\} + \{12, 16\} = \{\mathbf{18}, \mathbf{28}\} \end{aligned}$$

Discrete time system: Impulse Decomposition Property

- Any **discrete-time signal** can be built by **adding** together shifted impulses (**delta functions**) **multiplied** by the **corresponding signal values**.
- Mathematically, any discrete time signal, $x[n]$, can be represented by a **weighted sum of delayed impulses**, i.e.,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

where, $\delta[n - k]$ is a **shifted unit impulse** or **delayed delta function**.

- The **unit impulse** $\delta[n]$ is defined as:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Sample Example

Example: Suppose $\mathbf{x}[n] = \{3, 4, 5\}$

i. e., $\mathbf{x}[n] = \{x[0] = 3, x[1] = 4, x[2] = 5\}$

- Then the decomposition can be written as

$$\mathbf{x}[n] = 3\delta[n] + 4\delta[n - 1] + 5\delta[n - 2]; \text{ where, } k=0, 1, 2$$

- for $\mathbf{n} = 0$, $3\delta[0] + 4\delta[-1] + 5\delta[-2]$
 $= 3(1) + 4(0) + 5(0) = 3$
- for $\mathbf{n} = 1$, $3\delta[1] + 4\delta[0] + 5\delta[-1]$
 $= 3(0) + 4(1) + 5(0) = 4$
- for $\mathbf{n} = 2$, $3\delta[2] + 4\delta[1] + 5\delta[0]$
 $= 3(0) + 4(0) + 5(1) = 5$
- Signal is successfully reconstructed $\{3, 4, 5\}$

Convolution in time-domain

- An important property of an LTI system is that it can be completely characterized by its **impulse response**.
- ***Impulse response** is defined as a system's response to a unit sample (or an **impulse signal**).*
- Given an LTI system, the output $y[n]$ for an input $x[n]$ is given by a sum of **weighted and delayed impulse responses**, i.e.,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] = x[n] * h[n]$$

- where $h[n]$ is the **impulse response** of the system
- Each input sample $x[k]$ excites the system's impulse response $h[n]$.
- This is referred to as the convolution of $x[n]$ with $h[n]$.

- Once we know $h(t)$, we can find the output of the system for **any input** using **convolution**:

$$y(t) = x(t) * h(t)$$

- The impulse response contains all the information about how the system behaves.
- **Convolving the input signal $x[n]$ with the system's impulse response $h[n]$** is referred to as **filtering** the input – a fundamental process in speech production modelling, extensively used in almost every speech processing application.
- **Filtering** (Convolution with the system's impulse response):
 - Decompose the input signal into tiny impulses.
 - Replace each impulse with the corresponding scaled and shifted impulse response.

Convolution Properties

- **Commutative:** [whether we treat $x[n]$ as input or $h[n]$ as system, the output remains the same.]

$$x[n] * h[n] = h[n] * x[n]$$

- **Associative:** $x[n] * (h[n] * g[n]) = (x[n] * h[n]) * g[n]$

- **Distributive:** [Convolution distributes over addition.]

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

- **Shift Property:** If $x[n - n_0]$ is a shifted version of $x[n]$, then: $x[n - n_0] * h[n] = y[n - n_0]$; time-invariance preserved.

- **Impulse Property:** [Convolution with a unit impulse leaves the signal unchanged.]

$$x[n] * \delta[n] = x[n]$$

- **Scaling Property:** [Scaling the input scales the output by the same factor.]

$$(a \cdot x[n]) * h[n] = a \cdot (x[n] * h[n])$$

- **Finite Duration Property:** If $x[n]$ has length N_1 and $h[n]$ has length N_2 , then $y[n]$ has length $N_1 + N_2 - 1$.

Sample Example

Using the convolution operation, determine the output sequence $y[n]$ of an LTI system for the following input signal and impulse response:

$$x[n] = \{1, 2, 1\}, n = 0, 1, 2$$

$$h[n] = \{1, -1\}, n = 0, 1$$

Sol:

Input signal: $x[n] = \{1, 2, 1\}$ (at, $n = 0, 1, 2$)

Impulse response: $h[n] = \{1, -1\}$ (at, $n = 0, 1$)

$$\begin{aligned}\text{Output length, } \text{len}(y) &= \text{len}(x) + \text{len}(h) - 1 \\ &= 3 + 2 - 1 = 4\end{aligned}$$

$$i \rightarrow y[0] \quad y[1] \quad y[2] \quad y[3]$$

S1: Write the signal with time indices

n:	0	1	2
x[n]:	1	2	1
h[n]:	1	-1	

S2: Flip the impulse response

Original

After flip

n: 0	1	n: -1	0
h[n]: 1	-1	h[-n]: -1	1

S3: Find $y[0]$

$$\begin{array}{rcccc} n: & & \mathbf{0} & 1 & 2 \\ x[n]: & & & \mathbf{1} & 2 & 1 \\ h[-n]: & & -1 & \mathbf{1} & & \end{array}$$

$$\text{So, } y[0] = 1 \times 1 = 1$$

- **Find $y[1]$ (Slide the impulse response by 1 step)**

$$\begin{array}{rcccc} n: & & \mathbf{0} & 1 & 2 \\ x[n]: & & \mathbf{1} & 2 & 1 \\ h[-n]: & & -\mathbf{1} & 1 & & \end{array}$$

$$\text{So, } y[1] = (1 \times -1) + 2 \times 1 = -1 + 2 = 1$$

- **Find $y[2]$ (Slide the impulse response by 1 step)**

$$\begin{array}{rcccc} n: & 0 & 1 & 2 & \\ x[n]: & 1 & 2 & 1 & \\ h[-n]: & & -1 & 1 & \end{array}$$

$$\text{So, } y[2] = 2 \times -1 + 1 \times 1 = -2 + 1 = -1$$

- **Find $y[3]$ (Slide the impulse response by 1 step)**

$$\begin{array}{rcccc} n: & 0 & 1 & 2 & \\ x[n]: & 1 & 2 & 1 & \\ h[-n]: & & & -1 & 1 \end{array}$$

$$\text{So, } y[3] = 1 \times -1 = -1$$

- **Final Output: $y[n] = \{1, 1, -1, -1\}$**

Practice Problems

1. A discrete-time LTI system has the
input signal $x[n] = \{2, 1, 2, 4, 3\}$, $n=0, 1, 2, 3, 4$; and the
impulse response $h[n] = \{1, -1, 2\}$, $n=0, 1, 2$.
Using the convolution operation, determine the output
sequence $y[n]$.
2. A discrete-time LTI system has the
input signal $x[n] = \{1, 3, 2, 1\}$, $n=0, 1, 2, 3$ and the
impulse response $h[n] = \{2, -1\}$, $n=0, 1$.
Using the convolution operation determine the output
sequence $y[n]$.

Practice Problems (contd...)

1. $x[n] = \{1, 2, 1, 3\}$, $h[n] = \{1, 0, -1\}$. Determine output length and compute all $y[n]$.
2. $x[n] = \{2, -1, 3\}$, $h[n] = \{1, -2, 1\}$. Compute $y[n]$ and verify commutative property.
3. Express $x[n] = \{3, 4, 5\}$ as a sum of impulses and use convolution with $h[n] = \{1, 2\}$ to find $y[n]$.
4. $x[n] = \{1, 2\}$, $h_1[n] = \{1, 1\}$, $h_2[n] = \{1, -1\}$.
Show $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$.

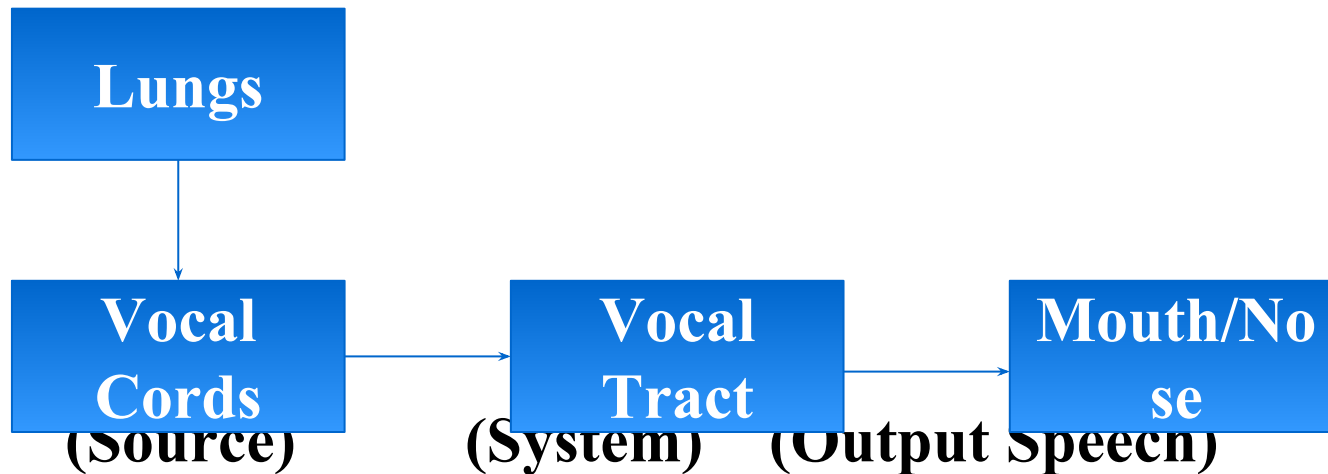
Limitations of Time-Domain Convolution

- **Computationally intensive:** Convolution requires repeated shifting, multiplication, and addition of signal samples, making it time-consuming for long signals.
- **Difficult to understand filter behavior:** It is not easy to see how a filter affects different frequency components by looking only at the time-domain signal.
- **No direct frequency information:** The time-domain representation does not clearly show important characteristics such as frequency content, bandwidth, or resonant frequencies.

- **Limited system analysis:** Properties like poles, zeros, and frequency response cannot be easily observed from the time-domain representation alone.
- **Complex signals are hard to interpret:** Speech and music signals often appear irregular in the time domain, making it difficult to identify harmonics, formants, or other frequency components.
- **Time-domain convolution** helps us determine the output of an LTI system for a given input, but it provides limited information about the internal characteristics of the system.
 - To better understand and represent how the speech production system behaves, we use Pole-Zero Modeling, which describes the vocal tract using a mathematical system model.

Pole-Zero Modeling

- **Pole-zero modeling** is a mathematical technique used to represent how the human speech production system modifies the sound produced by the vocal cords.
- It is widely used in speech processing, speech coding, and speech recognition.
- **Speech Production System**



Why Pole-Zero Modeling?

- In digital speech processing, **Vocal Tract** is considered as linear system.
- The **vocal tract** shapes the sound produced through the vocal cords by doing the physical movements of tongue, lips, jaw etc.,
- Instead of describing all these physical movements, DSP represents the **vocal tract** using a transfer function known as the **Pole-Zero Model**.
- Mathematically, the speech system can be represented as below:

Excitation Signal ($\mathbf{x}[\mathbf{n}]$) $\rightarrow$ Vocal Tract System ($\mathbf{H}(\mathbf{z})$) $\rightarrow$ Speech ($\mathbf{y}[\mathbf{n}]$)

- The vocal tract is represented as transfer function as below:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_M z^{-M}} = \frac{(z - z_1)(z - z_2) \dots}{(z - p_1)(z - p_2) \dots}$$

Where,

$Y(z)$, Z-transform of the output signal and contains Zeros

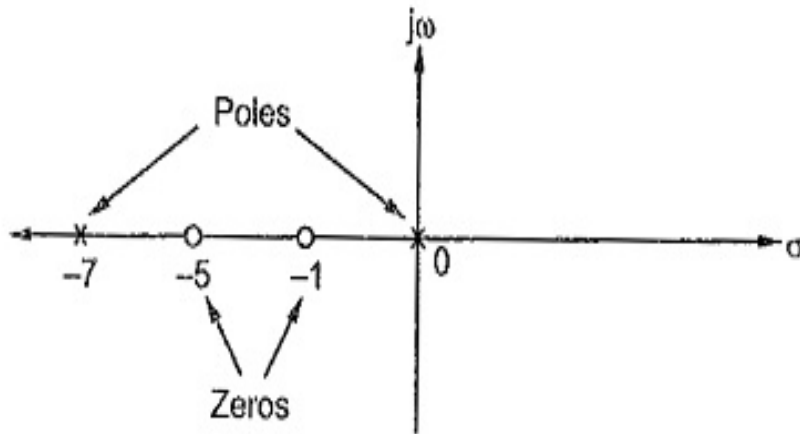
$X(z)$, Z-transform of the output signal and contains Poles

- A **Pole** is a value that **increases or amplifies** certain frequencies. In speech, Poles represent the resonances (formants) of the vocal tract.
- A **Zero** is a value that **reduces or cancels** certain frequencies.

Pole-Zero Plot

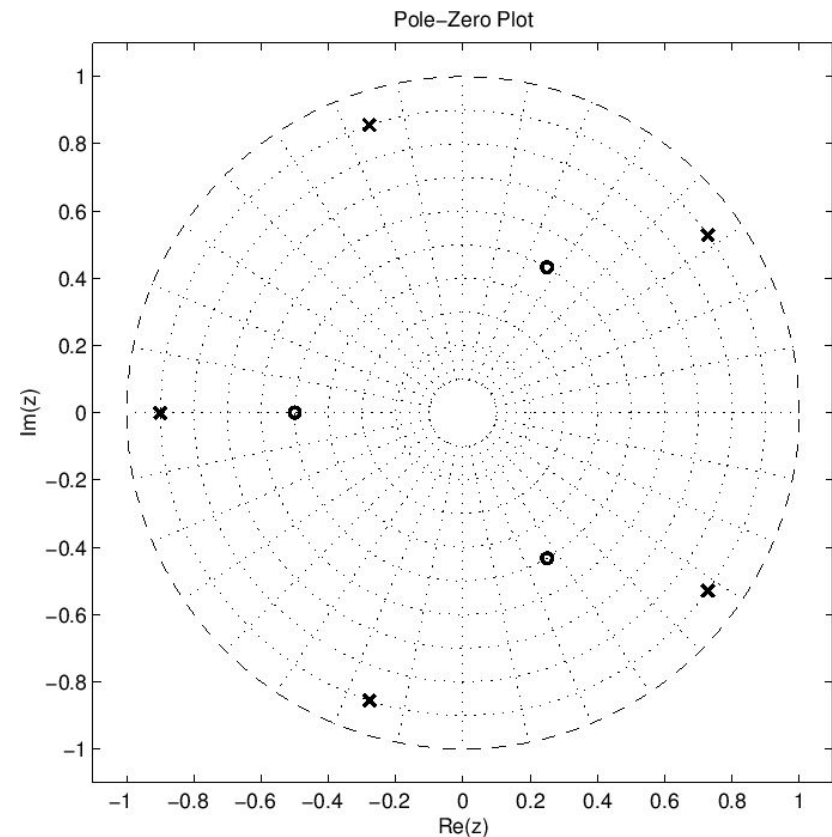
- A **pole-zero plot** is plotted in the plane of a complex frequency domain, showing the location of the **poles and zeros** of the transfer function.
- **Continuous-time systems** use the **Laplace transform** (plotted in the *s-plane*) whereas, **Discrete-time systems** use the **Z-transform** (plotted in the *z-plane*).
- By convention, the **poles** of the system are indicated in the plot by an **X** while the **zeros** are indicated by a **circle or O**.

contd...



s-plane pole-zero plot for continuous time-domain

z-plane pole-zero plot for discrete time-domain



Example-1

Determine poles and zeros for the given Let's consider the transfer function:

$$H(z) = \frac{z + 2}{z^2 + \frac{1}{4}}$$

Sol:

For Zeros, set the numerator to zero.

$$z + 2 = 0 \text{ i. e., } z = -2$$

So the system has one zero at $z = -2$.

i.e., The system completely suppresses the response at $z = -2$

- **For Poles, set the denominator to zero.**

$$z^2 + \frac{1}{4} = 0 \text{ i.e., } z^2 = -\frac{1}{4}$$

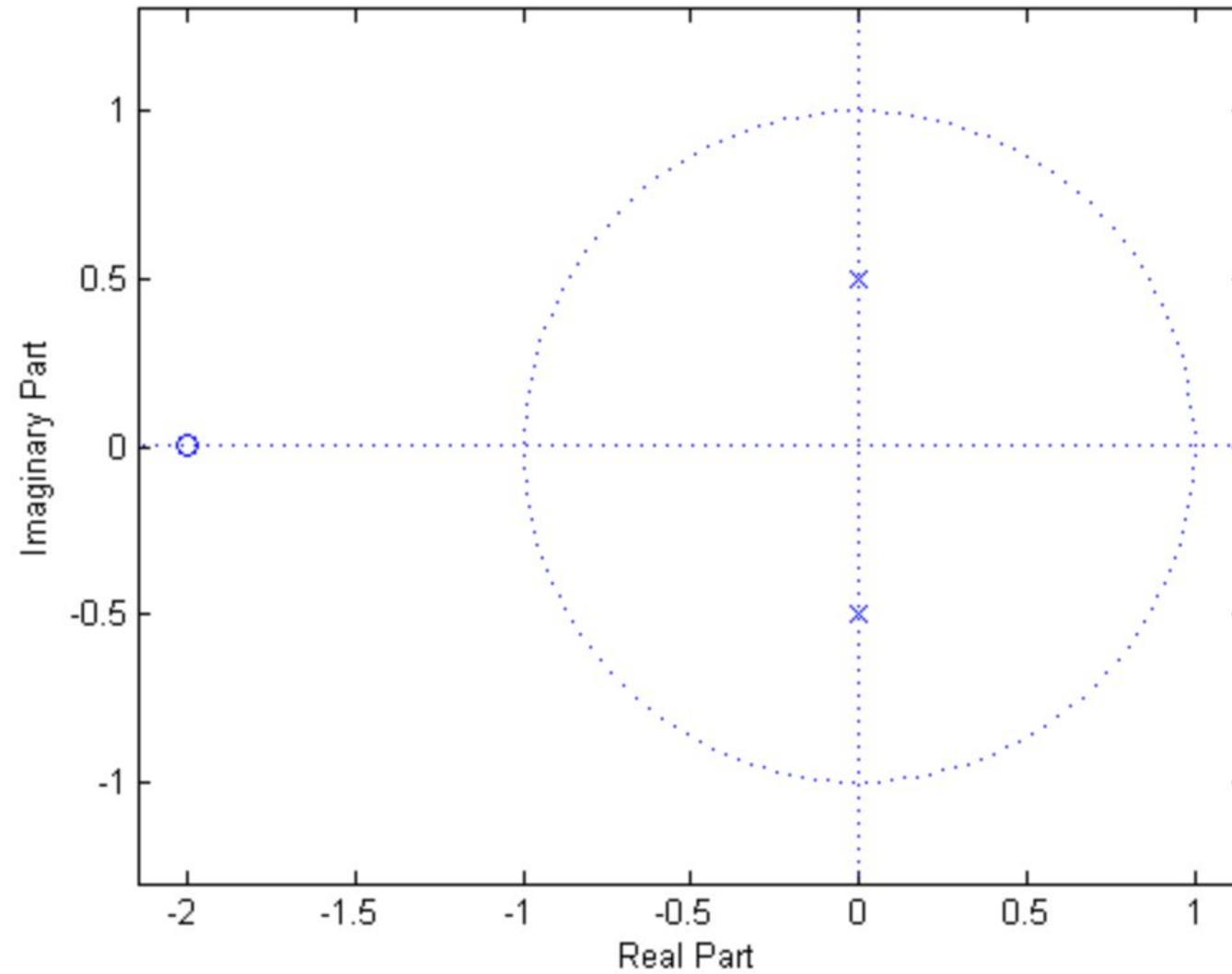
$$z = \pm \sqrt{-\frac{1}{4}}$$

$$\text{Since } \sqrt{-1} = j; \text{ we get } z = \pm \frac{j}{2}$$

Thus the system has two poles: $+\frac{j}{2}$ and $-\frac{j}{2}$

i.e., This means the system's response becomes very large (theoretically infinite) at $z = \pm \frac{j}{2}$

contd...



Example-2

- Determine poles and zeros for the given Let's consider the transfer function:

$$X(z) = \frac{z^2(z + 1.2)}{(z - 0.5 + j0.7)(z - 0.5 - j0.7)(z - 0.8)}$$

Sol: For Zeros, set the numerator to zero.

$$z^2(z + 1.2) = 0$$

Now, here we have two components z^2 and $(z + 1.2)$. Lets deal with them separately.

$$z^2 = 0$$

$z = 0$ (taking the square root on both the sides)

Hence, $z = (0,0)$; these are called repeated zeros.

$$\text{Now, } z + 1.2 = 0$$

For Poles, set the denominator to zero.

$$(z - 0.5 + j0.7)(z - 0.5 - j0.7)(z - 0.8)$$

So, every factor is set to zero.

$$(z - 0.5 + j0.7) = 0$$

$$z = 0.5 - j0.7$$

So, pole is at (0.5, -0.7)

Again,

$$(z - 0.5 - j0.7) = 0$$

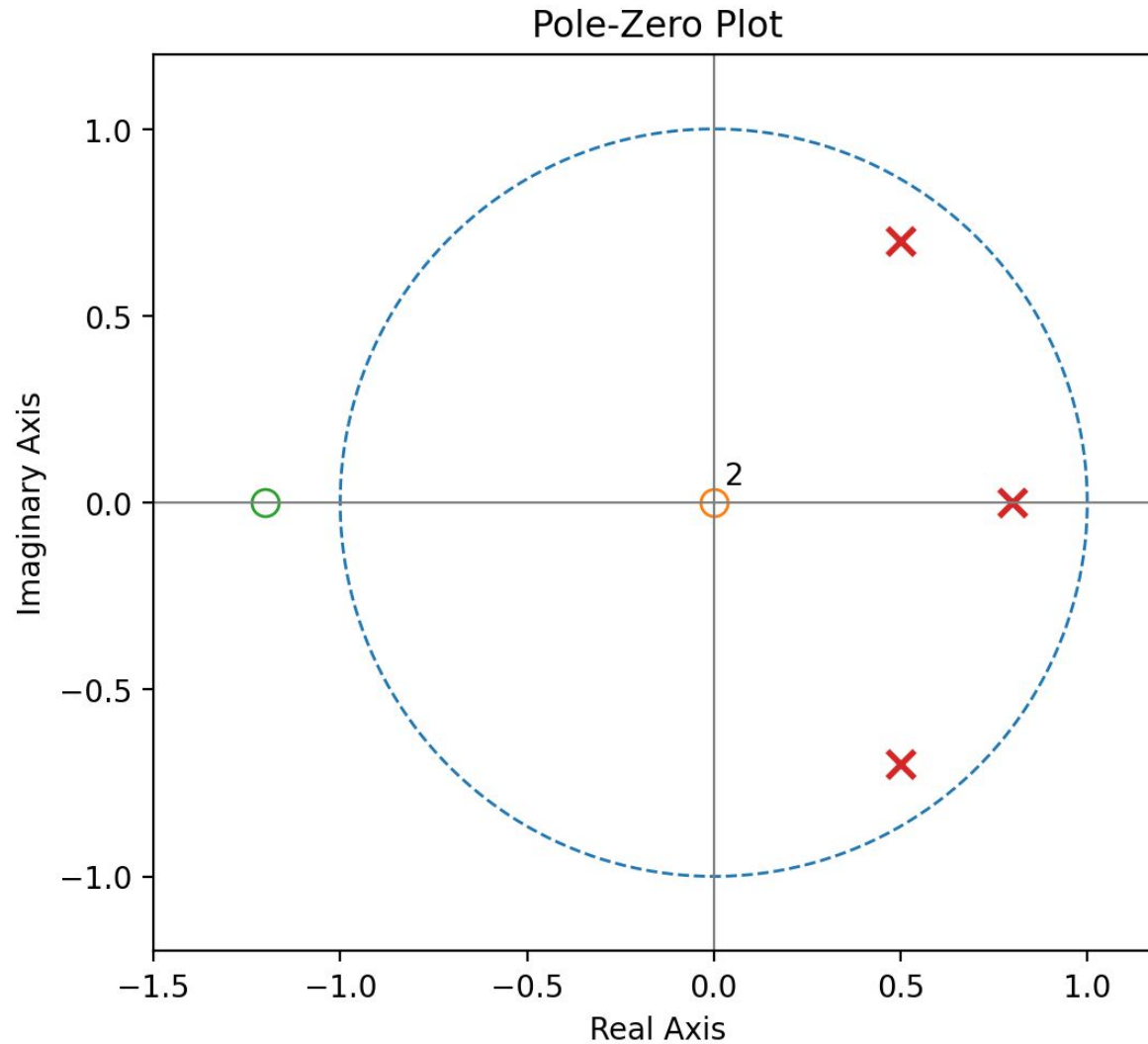
$$z = 0.5 + j0.7 ; \text{ So, pole is at (0.5, +0.7)}$$

Finally, $z - 0.8 = 0$

$$z = 0.8$$

So, pole is at (0.8, 0)

Pole-Zero Plot is as below:



Significance of Poles and Zeroes

Pole-zero plot shows stability and resonance of the system.

- In z-plane, **poles** must lie inside the unit circle (discrete-time)[in s-plane left half-plane (continuous-time)]. This indicates **bounded response** that makes the system stable.
- **Poles** lead to sharp peaks in frequency response (resonance).
- **Zeros** represent frequencies where the system output becomes zero.
 - The frequency response can be shaped by cancelling certain frequency components.
- The **distance of poles and zeros** from the unit circle determines how **strong or weak** these effects are.
- Poles and zeroes together shapes system's filtering behaviour. Pole-zero placement is the **design backbone of digital filters.**

Pole-Zero Modeling for Speech

- Speech can be modeled as:

$s[n] = e[n] * h[n]$ where $e[n]$: **excitation** (voiced source like glottal pulses, or unvoiced noise) and $h[n]$: vocal tract filter.

- The vocal tract filter is represented by a **transfer function** with **poles** (resonances) and **zeros** (anti-resonances).
- **Poles** correspond to **formants** (resonant frequencies of the vocal tract).
 - **Example:** vowel /a/ has formants around 700 Hz, 1200 Hz, 2500 Hz.
- These appear as poles near the unit circle in the z-plane.
- **Zeros** model **anti-resonances**, such as nasal sounds or fricatives.

- Determine the poles and zeros for

$$H(z) = \frac{(z - 2)(z + 1)}{z - 0.8}$$

- Find the poles and zeros of

$$H(z) = \frac{z(z + 2)}{(z - 0.6)(z + 0.4)}$$

- Determine the poles and zeros of

$$H(z) = \frac{z^2(z + 1)}{(z - 0.8)(z + 0.5)}$$

- Determine poles and zeros for

$$H(z) = \frac{(z + 0.5 + j0.5)(z + 0.5 - j0.5)}{(z - 0.6 + j0.3)(z - 0.6 - j0.3)}$$